

Total No. of Questions : 8]

SEAT No. :

PD4659

[Total No. of Pages : 2

[6404]-165

B.E. (Information Technology)

DEEP LEARNING

(2019 Pattern) (Semester-VII) (414443)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6 and Q.7 or Q.8.
- 2) Assume suitable data if necessary.
- 3) Neat diagrams must be drawn wherever necessary.
- 4) Figures to the right indicate full marks.

Q1) a) Explain various types of Recurrent Neural Networks with architecture diagram and applications. [9]

b) Differentiate between RNN and FFNN. [9]

OR

Q2) a) Explain Recursive Neural Networks with diagram and applications. [9]

b) Explain Long Short-Term Memory Network with suitable architectural diagram and application areas. [9]

Q3) a) What are encoders? Explain undercomplete auto encoder. [8]

b) What is regularized auto encoder? Where are they used? [9]

OR

Q4) a) Write a short note on-Stochastic autoencoders and decoders. [8]

b) Differentiate between denoising auto encoder and contractive auto encoder. [9]

Q5) a) Write a short note on - DenseNet [9]

b) What is domain adaption? Explain with suitable example. [9]

OR

P.T.O.

- Q6)** a) What is transfer learning? Explain with suitable example. [9]
b) What is greedy layer-wise pre-training network? [9]

- Q7)** a) Explain how deep learning is used in recommender systems? [8]
b) Explain deep learning-based framework for NLP. [9]

OR

- Q8)** a) Explain the role of CNNs as an image classifier. [8]
b) How deep learning is used in social network analysis? [9]

